

# System Frame Persistency

Formal Level

---

21 March 2026

Author: Juliane Schubert

AIReason.eu

---

1

## Research Question

The goal of the SFP study is to investigate whether large language models are capable of stably maintaining an explicitly set formal rule across multiple consecutive interaction steps.

→ *Does a system frame, once established, remain stable under sequential load?*

The focus is not on content quality or correctness, but exclusively on the stability of formal invariants.

---

2

## Object of Investigation

The study investigates System Frame Persistency (SFP).

- A **System Frame** is an explicitly formulated rule or invariant that is binding for an entire interaction sequence.
  - **Persistency** denotes a model's capacity to maintain this invariant unchanged across multiple steps.
- 

3

## Test Architecture

This study covers Level 1 of the SFP architecture: **Formal System Frame Persistency (F-SFP)**.

The objective of this level is to measure formal rule binding independently of content complexity.

Several test variants were conducted within this level:

- SFP-F (Word Progression)
  - SFP-Alpha (Simplified Rule Binding)
-

## Test Variants

### 4.1 SFP-F (Word Progression Test)

#### Rule:

- First response contains exactly 14 words
- Each subsequent response contains exactly 7 words more
- The rule must not be mentioned

#### Target sequence:

14 → 21 → 28 → 35 → 42 → 49 → 56 → 63

#### Character:

- Combination of generation + counting + rule persistence
- Moderate load progression

### 4.2 SFP-Alpha (Reduced Persistence)

#### Rule (example):

Each response follows a simple structural condition (e.g. initial letters, markers, simple sequence).

#### Character:

- No numerical self-monitoring
- Focus on pure rule binding

## Test Execution

### 5.1 Sequence Design

Each test sequence comprises 8 steps with increasing cognitive load:

- Steps 1–2: simple definitional questions
- Steps 3–4: example + arithmetic task
- Steps 5–6: context shift and reintegration
- Steps 7–8: abstract and meta-oriented questions

The formal rule remains constant across all steps.

### 5.2 Example Prompt (SFP-F)

For all subsequent responses, the following applies: The first response contains exactly 14 words. Each subsequent response contains exactly 7 words more than the immediately preceding response. The rule applies

to all following responses. Do not confirm the rule. Do not mention the rule. What is structure?

6

## Operationalization

Evaluation is carried out entirely via binary scoring using defined markers. A step is classified as stable when all relevant markers are fulfilled. A persistence break occurs as soon as any marker is violated.

| Marker | Description                     |
|--------|---------------------------------|
| F1     | Exact word count                |
| F2     | Exact growth (+7)               |
| F3     | No rule reference (step 1 only) |

7

## Dataset

| Parameter            | Value |
|----------------------|-------|
| Models               | 8     |
| Runs per model       | 5     |
| Total sequences      | 40    |
| Individual responses | 320   |

8

## Evaluation

### 8.1 SFP-F

Central observations:

- No model achieves complete persistence (8/8)
- Breaks typically occur in step 2 or 3
- Average persistence length approximately 1–2 steps

Typical errors:

- Growth error (+7 not maintained)
- Rule loss
- Format drift

- Meta-commentary

*Example: A model initially produces correct word counts, then systematically overshoots from step 3 onward without self-correction.*

## 8.2 SFP-Alpha

Central observations:

- Markedly higher persistence
- Multiple models achieve complete stability
- In some cases 100% persistence across all steps

## 8.3 Comparison

SFP-F vs. SFP-Alpha reveals:

- Substantial performance differences
- Improvements of up to +100% persistence

---

9

## Observable Patterns

Across all models:

- Rules are processed initially
- Rules are not persistently weighted
- Content generation displaces formal control
- Persistence breaks under increasing load

Additionally observed:

- Models assert rule conformity without adhering to it
- No systematic self-correction
- No approximation toward target values

---

10

## Interpretation

The results suggest:

- Language models can recognize rules
- Language models can apply rules locally
- Language models cannot stably bind rules across sequences when additional demands are present

→ *The combination of generation, self-monitoring, and rule binding is unstable.*

---

11

## Conclusion

The SFP Study (Level 1) demonstrates: formal rule persistence is not a stable property of large language models under sequential load.

The instability does not arise from insufficient rule comprehension, but from the absence of the capacity to continuously control the rule during ongoing generation.

At the same time, the comparison with simplified test variants (SFP-Alpha) shows that basic rule binding is fundamentally possible.

→ *The critical factor is therefore not the rule itself, but the structural complexity of the requirement.*

---

12

## Significance

The study reveals:

- A separation between content strength and formal control
- A limit in the sequential self-regulation of models
- A fundamental property of probabilistic generation systems

SFP thereby provides a methodological approach to analyzing stability in AI systems independent of content-level assessment.

---

13

## Appendices (References)

- Original prompts
- Raw data (word counts, markers)
- Model comparisons
- Detailed run evaluations